TUTORIAL 7

Rank, Row and Column Spaces, and Measurement Systems

MOTIVATION AND GUIDING IDEA

Motivation. In $Ax = b$, the column space determines attainable measurements, while the kernel determines undetectable changes in the state. Distinguishing these roles allows existence and uniqueness to be tested before solving explicitly.

NOTATION AND CONVENTIONS

- $\mathbb{F}^{m\times n}$ is the space of matrices with m rows and n columns over the field $\mathbb{F}$; A^T denotes the transpose.
- $\mathcal{R}(A) = \text{Row}(A) \subseteq \mathbb{F}^n$ is the row space; $\mathcal{C}(A) = \text{Col}(A) \subseteq \mathbb{F}^m$ is the column space. Identify a row with its transposed column when matrix multiplication requires it.
- $N(A) = \ker A = \{x \in \mathbb{F}^n : Ax = 0\}$ is the kernel; $\text{nullity}(A) = \dim \ker A$ is the nullity; $\text{rank}(A) = \dim \text{Col}(A) = \dim \text{Row}(A)$ is the rank.
- span S consists of all finite linear combinations of vectors in S; e_j is the standard basis vector with entry 1 at index j and 0 elsewhere.
- $x_0 + \ker A = \{x_0 + v : v \in \ker A\}$ is an affine space, a translate of the kernel. Its affine dimension is $\dim \ker A$; it need not be a vector subspace.

DEFINITION 7.1

Let $A \in \mathbb{F}^{m\times n}$, and define $T_A : \mathbb{F}^n \to \mathbb{F}^m$ by $T_A(x) = Ax$. Then $\text{Im}\, T_A = \text{Col}(A)$ and $\ker T_A = N(A)$, so

$$\text{rank}(A) + \dim \ker A = n.$$

The right-hand side is the dimension of the **domain**. The system $Ax = b$ is solvable if and only if $b \in \text{Col}(A)$. If x_0 is one solution, **all** solutions form $x_0 + \ker A$. An existing solution is therefore unique if and only if $\ker A = \{0\}$. The nonzero echelon rows form a basis of the row space, but a column-space basis must use the **original** columns at the pivot indices.

Created by Chen Gadi

 © 2026 Chen Gadi

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Distinguish the row space, column space and null space, and use rank and nullity to quantify information and freedom in a system. The measurement examples apply to engineering systems in many disciplines.

7.1 Row and column spaces and the rank theorem

RECALL AND TOOLS

The **row space** $\mathcal{R}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^n$ spanned by its rows. .

- Elementary row operations preserve the row space of a matrix.

PROPOSITION 7.1

- **Claim 1.** If $A, B \in M_{m\times n}(\mathbb{F})$ are row equivalent, then $\mathcal{R}(A) = \mathcal{R}(B)$.
- **Claim 2:** If A is in row echelon form, its nonzero rows form a basis of $\mathcal{R}(A)$, (and are therefore linearly independent in $\mathbb{F}^n$).

RECALL AND TOOLS

The **column space** $\mathcal{C}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^m$ spanned by its columns. .

- Row reduction **does not in general preserve the column space**; only its **dimension** is preserved.
- $\mathcal{C}(A) = \mathcal{R}(A^T)$ when a row vector is identified with its transposed column vector.

PROPOSITION 7.2

The rank theorem: Every matrix A satisfies $\dim \mathcal{C}(A) = \dim \mathcal{R}(A)$. This common dimension is called the **rank of the matrix**.

RECALL AND TOOLS

- The row and column spaces need not be the same subspace. For a rectangular matrix they even lie in different ambient spaces, $\mathbb{F}^n$ and $\mathbb{F}^m$, respectively; their dimensions are nevertheless equal.
- Equivalently, rank$A$ is the number of pivots in the reduced row echelon form.
- For $A \in \mathbb{R}^{m\times n}$, $\text{rank}(A) \leq \min(m, n)$.
- A square matrix $A \in \mathbb{R}^{n\times n}$ is invertible if and only if $rank(A) = n$
- $rank(A \cdot B) \leq min\{rank(A), rank(B)\}$
- $rank(A + B) \leq rank(A) + rank(B)$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.1

$$A = \begin{pmatrix} 2 & 6 & -4 \\ 1 & 3 & -2 \end{pmatrix}.$$

Please find a basis and dimension for the row space and column space of A.

WORKED SOLUTION

Swap the rows and subtract twice the first row from the second. The column-space basis is taken from the original matrix.

$$A \sim \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & -2 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$\mathrm{Row}(A) = \mathrm{span}\{(1,3,-2)\}, \qquad \mathrm{Col}(A) = \mathrm{span}\{(2,1)^T\}.$$

$$\mathrm{Col}_2(A) = 3\,\mathrm{Col}_1(A), \qquad \mathrm{Col}_3(A) = -2\,\mathrm{Col}_1(A), \qquad \dim \mathrm{Row}(A) = \dim \mathrm{Col}(A) = 1.$$

EXERCISE 7.2

Let $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{n\times m}$ satisfy $AB = I_m$. Please find the ranks of A and B.
What can be concluded about m and n?

WORKED SOLUTION

The rank inequalities give $\mathrm{rank}(AB) = \mathrm{rank}(I_m) = m \le \mathrm{rank}(A) \le m$.
Thus $\mathrm{rank}(A) = m$. For B, $m = \mathrm{rank}(AB) \le \mathrm{rank}(B) \le m$, so $\mathrm{rank}(B) = m$ as well.
Since $\mathrm{rank}(A) = m \le \min(m,n)$, we obtain $m \le n$.
The matrices are not assumed square, so one cannot conclude that B is an inverse of A.

7.1.1 Reducing spanning sets and extending independent sets to bases in $\mathbb{F}^n$

Every spanning set contains a basis, and every linearly independent set extends to a basis.

Reducing a spanning set to a basis

MOTIVATION AND GUIDING IDEA

1. **Method A:** We want a basis consisting of vectors from the original set (as in the preceding exercise).
   - Form a matrix whose **columns** are the vectors of S and reduce it to row echelon form.
   - For each pivot column, choose the corresponding **original column** as a basis vector. These columns are linearly independent.
   - The column space is not preserved by row operations; therefore the columns must be taken from the original matrix.
2. **Method B:** Find a **new** linearly independent set spanning the same space.
   - Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
   - Its nonzero echelon rows form a basis.

Created by Chen Gadi

 © 2026 Chen Gadi

- Row operations preserve the row space, so the basis may be taken after reduction.

Extending a linearly independent set to a basis

MOTIVATION AND GUIDING IDEA

- Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
- For each nonpivot column j, adjoin the standard basis vector e_j to S. To justify this construction, first replace the original rows by the echelon rows, preserving their span. At the pivot indices, the added standard basis vectors are zero, whereas the echelon rows give a triangular matrix with nonzero diagonal. Hence a vanishing linear combination first forces the coefficients of the echelon rows to be zero, and then those of the added standard basis vectors. Since the original rows are independent, replacing the echelon rows back by the originals preserves the basis property.

EXERCISE 7.3

$$S = \{(1,2,0,1,-1)^T, (2,4,1,3,0)^T, (1,2,1,2,2)^T\} \subseteq \mathbb{R}^5.$$

Please prove that S is linearly independent and please extend it to a basis of $\mathbb{R}^5$.

WORKED SOLUTION

The three nonzero echelon rows prove independence. Pivot columns are $1, 3, 5$. Adding e_2, e_4 gives five independent rows: the pivot columns first force the three echelon-row coefficients to vanish, and columns $2, 4$ then force the remaining two coefficients to vanish.

$$\begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 4 & 1 & 3 & 0 \\ 1 & 2 & 1 & 2 & 2 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$B = S \cup \{e_2, e_4\}.$$

EXERCISE 7.4

$$V = \operatorname{span}\{(1,3,2,-1)^T, (2,7,5,0)^T, (0,1,1,2)^T\} \subseteq \mathbb{R}^4.$$

Please find a basis and dimension for V and please extend that basis to a basis of $\mathbb{R}^4$.

WORKED SOLUTION

The first two coordinates show that the two vectors in B are independent. They span all three given vectors. The final determinant verifies the extension.

$$(2,7,5,0)^T = 2(1,3,2,-1)^T + (0,1,1,2)^T.$$

$$B = \{(1,3,2,-1)^T, (0,1,1,2)^T\}, \qquad \dim V = 2.$$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.4 CONTINUED

$$\det \begin{pmatrix} 1 & 3 & 2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 1, \qquad D = B \cup \{e_3, e_4\}.$$

EXERCISE 7.5 EXAM QUESTION

Work over $\mathbb{Z}_3 = \{0, 1, 2\}$, with all arithmetic modulo 3. For each $\omega \in \mathbb{Z}_3$, define

$$A(\omega) = \begin{pmatrix} 1 & \omega & 1 \\ \omega & 1 & 1 \\ 1 & 1 & \omega \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \qquad c = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Please determine the values of ω for which $b \in \mathrm{Col}(A(\omega))$, and please find all solutions of $A(\omega)x = b$ in those cases.
2. Please determine the values of ω for which both $b \in \mathrm{Col}(A(\omega))$ and $c \in \mathrm{Col}(A(\omega))$ hold.

WORKED SOLUTION

All operations below are modulo 3. For $\omega = 0, 2$ the matrix is invertible, so the displayed solutions for b are unique. For $\omega = 1$ its column space consists of equal-entry vectors, so the system with b is inconsistent. Both parts therefore have the answer $\omega \in \{0, 2\}$. The last products also give c in each of these two cases.

$$A(0) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A(1) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad A(2) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

$$\det A(\omega) = -\omega^3 + 3\omega - 2, \qquad \det A(0) = 1, \quad \det A(2) = 2 \quad \text{in } \mathbb{Z}_3.$$

$$A(0)\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = b, \qquad A(2)\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = b.$$

$$\mathrm{Col}(A(1)) = \{(a, a, a)^T : a \in \mathbb{Z}_3\}, \qquad b \notin \mathrm{Col}(A(1)).$$

$$A(0)\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = c, \qquad A(2)\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = c.$$

7.2 The null space and nullity of a matrix

DEFINITION 7.2

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m \times n}$.
The **null space of** A, also called its *kernel*, is the set of column vectors

$$N(A) := \{x \in \mathbb{F}^n \mid Ax = 0\},$$

Created by Chen Gadi

 © 2026 Chen Gadi

that solve the homogeneous system $Ax = 0$.

- $N(A)$ is a vector subspace of $\mathbb{F}^n$.
- Its dimension is the **nullity** of A, denoted

$$\dim N(A) = \text{null}(A).$$

PROPOSITION 7.3

For $A \in \mathbb{F}^{m\times n}$,

$$\text{rank}(A) + \text{null}(A) = n.$$

- A has n columns, each a vector in $\mathbb{F}^m$.
- Reduce A to reduced row echelon form.
- The number of pivot columns equals $\text{rank}(A)$. The columns of the *original matrix* at these pivot indices form a basis of $\text{Col}(A)$; row reduction itself need not preserve the column space.
- There are $n - \text{rank}(A)$ nonpivot columns. These correspond to free variables in $Ax = 0$, each contributing one independent parameter to its solution space. Consequently,

$$\dim N(A) = \text{null}(A) = n - \text{rank}(A).$$

- Therefore,

$$\text{rank}(A) + \text{null}(A) = \text{rank}(A) + \big(n - \text{rank}(A)\big) = n.$$

EXERCISE 7.6

$$A = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 2 \end{pmatrix}.$$

Please find a basis and dimension for the kernel, please give every solution of $Ax = 0$, and please check rank-nullity.

WORKED SOLUTION

The first two columns are pivot columns. Every solution arises from $t \in \mathbb{R}$; the nonzero spanning vector is a kernel basis.

$$x_1 - 3x_2 = 0, \quad x_2 + 2x_3 = 0, \quad x_3 = t \implies x_2 = -2t,\ x_1 = -6t.$$

$$\ker A = \text{span}\{(-6, -2, 1)^T\}, \qquad \text{rank } A + \dim \ker A = 2 + 1 = 3.$$

MOTIVATION AND GUIDING IDEA

Different questions require different spaces. The column space describes attainable right-hand sides. The null space describes changes in the input that leave the output unchanged. Rank and nullity measure these two kinds of freedom.

Created by Chen Gadi

 © 2026 Chen Gadi

7.3 Interpreting matrix-vector multiplication

DEFINITION 7.3

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m\times n}$. Write its columns as

$$A = \begin{pmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{pmatrix}, \qquad a_j \in \mathbb{F}^m.$$

Let $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{F}^n$.

1. **A linear combination of columns.** Matrix-vector multiplication is defined by

$$Av = v_1 a_1 + \cdots + v_n a_n.$$

Thus Av is the linear combination of the columns of A with coefficients $v_1, \dots, v_n$. It follows that

$$\{Av \mid v \in \mathbb{F}^n\} = \text{span}\{a_1, \dots, a_n\} = \text{Col}(A).$$

2. **A linear map.** The function

$$T_A : \mathbb{F}^n \to \mathbb{F}^m, \qquad T_A(v) = Av,$$

is linear: for all $u, v \in \mathbb{F}^n$ and $\lambda \in \mathbb{F}$,

$$T_A(u+v) = A(u+v) = Au + Av,$$

$$T_A(\lambda v) = A(\lambda v) = \lambda(Av).$$

Thus multiplication by A is a linear map, and its columns describe its action on the standard basis of $\mathbb{F}^n$:

$$T_A(e_j) = a_j.$$

Created by Chen Gadi

EXERCISE 7.7

Let $\mathbb{F}$ be a field and $B \in \mathbb{F}^{n\times m}$. Define

$$J := \{Bv \mid v \in \mathbb{F}^m\} \subseteq \mathbb{F}^n.$$

Please show that

$$J = \text{Col}(B).$$

WORKED SOLUTION

Denote the columns of B by

$$B = \begin{pmatrix} | & & | \\ b_1 & \dots & b_m \\ | & & | \end{pmatrix},$$

so that $b_1, \dots, b_m \in \mathbb{F}^n$ are its column vectors.

 © 2026 Chen Gadi

EXERCISE 7.7 CONTINUED

Now let

$$v = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{pmatrix} \in \mathbb{F}^m.$$

By the definition of matrix-vector multiplication,

$$Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m.$$

We prove both inclusions.

1. First, $J \subseteq \mathrm{Col}(B)$.

   Let $y \in J$. Then $y = Bv$ for some $v \in \mathbb{F}^m$.

   Every such vector is a linear combination of $b_1, \ldots, b_m$:

   $$y = Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m$$

   for suitable scalars $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$.

   Thus y belongs to their span, namely

   $$y \in \mathrm{span}\{b_1, \ldots, b_m\} = \mathrm{Col}(B).$$

   This proves $J \subseteq \mathrm{Col}(B)$.

2. Conversely, $\mathrm{Col}(B) \subseteq J$.

   Let $w \in \mathrm{Col}(B)$. There are scalars $\eta_1, \ldots, \eta_m \in \mathbb{F}$ such that

   $$w = \eta_1 b_1 + \cdots + \eta_m b_m.$$

   Define

   $$v := \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_m \end{pmatrix} \in \mathbb{F}^m.$$

   The preceding calculation gives

   $$Bv = \eta_1 b_1 + \cdots + \eta_m b_m = w.$$

   Hence $w = Bv$ for some $v \in \mathbb{F}^m$, so

   $$w \in J.$$

   This proves $\mathrm{Col}(B) \subseteq J$.

The two inclusions yield

$$J = \mathrm{Col}(B),$$

as required.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.8

Let $\mathbb{F}$ be a field and let $A, B \in \mathbb{F}^{n\times n}$ satisfy

$$AB = 0.$$

Please show that

$$\text{Col}(B) \subseteq N(A),$$

Every column of B, and every linear combination of its columns, is therefore annihilated by A.

WORKED SOLUTION

Use the definitions of column space and null space.
It suffices to prove that $Av = 0$ for every $v \in \text{Col}(B)$.
By the definition of column space, $v \in \text{Col}(B)$ implies that some $w \in \mathbb{F}^n$ satisfies

$$v = Bw.$$

Compute

$$Av = A(Bw) = (AB)w.$$

Since $AB = 0$,

$$(AB)w = 0w = 0,$$

and therefore

$$Av = 0.$$

We have proved that $Av = 0$ for every $v \in \text{Col}(B)$, or equivalently,

$$v \in N(A).$$

Consequently,

$$\text{Col}(B) \subseteq N(A),$$

as required.

Created by Chen Gadi

7.4 Exercises

EXERCISE 7.9

Let V be a vector space over a field $\mathbb{F}$, and let $S, R \subseteq V$ be arbitrary subsets, not necessarily finite.
Please write

$$\text{span}(S) = \left\{ \sum_{i=1}^{k} \alpha_i s_i \;\middle|\; k \in \mathbb{Z}_{\geq 0},\ \alpha_i \in \mathbb{F},\ s_i \in S \right\},$$

The empty sum is the zero vector, so $\text{span}(\varnothing) = \{0\}$. Define $\text{span}(R)$ and $\text{span}(S \cup R)$ similarly.
Please show that, for all $S, R \subseteq V$,

$$\text{span}(S) + \text{span}(R) = \text{span}(S \cup R).$$

 © 2026 Chen Gadi

EXERCISE 7.9 CONTINUED

WORKED SOLUTION

Write

$$U := \text{span}(S), \qquad W := \text{span}(R).$$

Recall that the sum $U + W$ is defined by

$$U + W := \{u + w \mid u \in U,\ w \in W\}.$$

We prove

$$\text{span}(S) + \text{span}(R) = \text{span}(S \cup R)$$

by establishing both inclusions:

$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R) \quad \text{and} \quad \text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

1. First prove

$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$

Let $v \in \text{span}(S) + \text{span}(R)$. Then $v = u + w$ for some $u \in \text{span}(S)$ and $w \in \text{span}(R)$:

$$v = u + w.$$

By the definition of $\text{span}(S)$, there are vectors $s_1, \ldots, s_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$ such that

$$u = \sum_{i=1}^{k} \alpha_i s_i.$$

Similarly, there are vectors $r_1, \ldots, r_\ell \in R$ and scalars $\beta_1, \ldots, \beta_\ell \in \mathbb{F}$ such that

$$w = \sum_{j=1}^{\ell} \beta_j r_j.$$

Therefore,

$$v = u + w = \sum_{i=1}^{k} \alpha_i s_i + \sum_{j=1}^{\ell} \beta_j r_j.$$

Every s_i and every r_j lies in $S \cup R$. Thus v is a finite linear combination of vectors in $S \cup R$.

By the definition of span,

$$v \in \text{span}(S \cup R).$$

Hence

$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$

2. For the reverse inclusion, prove

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.9 CONTINUED

Let $v \in \text{span}(S \cup R)$. There are $x_1, \ldots, x_t \in S \cup R$ and $\gamma_1, \ldots, \gamma_t \in \mathbb{F}$ such that

$$v = \sum_{i=1}^{t} \gamma_i x_i.$$

Partition the indices into disjoint sets: assign to S every vector belonging to S, and assign the rest to R. A vector in the intersection is counted only in the first set.

Define

$$I_S := \{i \in \{1, \ldots, t\} \mid x_i \in S\}, \qquad I_R := \{1, \ldots, t\} \setminus I_S.$$

Then v can be written as

$$v = \sum_{i \in I_S} \gamma_i x_i + \sum_{i \in I_R} \gamma_i x_i.$$

Set

$$u := \sum_{i \in I_S} \gamma_i x_i, \qquad w := \sum_{i \in I_R} \gamma_i x_i.$$

Observe that

- for $i \in I_S$, $x_i \in S$, so u is a linear combination of vectors in S; thus

$$u \in \text{span}(S).$$

- for $i \in I_R$, $x_i \in R$, so w is a linear combination of vectors in R; thus

$$w \in \text{span}(R).$$

We have obtained

$$v = u + w$$

with $u \in \text{span}(S)$ and $w \in \text{span}(R)$, hence

$$v \in \text{span}(S) + \text{span}(R).$$

This proves

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

The two inclusions give

$$\text{span}(S) + \text{span}(R) = \text{span}(S \cup R),$$

as required.

EXERCISE 7.10

$$U = \text{span}\{1, x, 2 + x\}, \qquad W = \{p \in \mathbb{R}_3[x] : p(-1) = p(2) = 0\}.$$

Please find bases and dimensions for $U + W$ and $U \cap W$.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.10 CONTINUED

WORKED SOLUTION

A polynomial of degree at most one with two distinct roots is zero. Thus the intersection is zero and the sum is direct. The four listed polynomials are independent by their degrees and span the ambient space. The intersection has the empty basis.

$$U = \operatorname{span}\{1, x\}, \qquad p \in W \iff p = (x+1)(x-2)(a+bx).$$

$$W = \operatorname{span}\{x^2 - x - 2,\ x^3 - x^2 - 2x\}, \qquad \dim W = 2.$$

$$U \cap W = \{0\}, \quad \dim(U \cap W) = 0, \qquad U + W = \mathbb{R}_3[x], \quad \dim(U+W) = 4.$$

$$B_{U+W} = (1, x, x^2 - x - 2, x^3 - x^2 - 2x).$$

EXERCISE 7.11

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & -1 & 5 \end{pmatrix}, \qquad y = (4, -1, 3, 10)^T.$$

Please find row-space and column-space bases and the number of dependent rows. Please decide whether $Mx = y$ is solvable and whether an existing solution is unique. Please replace the fourth entry of y by 9 and please find every solution.

WORKED SOLUTION

The first two rows are independent and the other two depend on them. The third column is twice the first minus the second, so the listed columns form a basis. The row relations give necessary right-hand-side conditions; they are sufficient because the first two equations can be solved for any two right-hand sides. The original right-hand side is inconsistent. After correction, setting $x_3 = t$ gives the displayed solutions. The one-dimensional kernel prevents uniqueness.

$$R_3 = R_1 + R_2, \qquad R_4 = 2R_1 - R_2, \qquad \operatorname{rank} M = 2.$$

$$B_{\text{row}} = \{(1, 0, 2), (0, 1, -1)\}, \quad B_{\text{col}} = \{(1, 0, 1, 2)^T, (0, 1, 1, -1)^T\}.$$

$$y \in \operatorname{Col} M \iff y_3 = y_1 + y_2, \quad y_4 = 2y_1 - y_2.$$

$$2 \cdot 4 - (-1) = 9 \neq 10, \qquad \ker M = \operatorname{span}\{(-2, 1, 1)^T\}.$$

$$x = (4, -1, 0)^T + t(-2, 1, 1)^T, \qquad t \in \mathbb{R}.$$

EXERCISE 7.12

$$m = Cq, \qquad C = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}.$$

Please find a column-space basis. Please prove that C is invertible and please calculate C^{-1} and q in terms of m.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.12 CONTINUED

WORKED SOLUTION

The matrix is triangular with diagonal entries 1, so it is invertible and its columns form a basis. Solving the equations in order gives the inverse formula.

$$\det C = 1, \qquad B_{\text{col}} = \{(1,2,-1)^T, (0,1,3)^T, (0,0,1)^T\}.$$

$$m_1 = q_1, \quad m_2 = 2q_1 + q_2, \quad m_3 = -q_1 + 3q_2 + q_3.$$

$$q_1 = m_1, \quad q_2 = m_2 - 2m_1, \quad q_3 = m_3 - 3m_2 + 7m_1.$$

$$C^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -3 & 1 \end{pmatrix}, \qquad CC^{-1} = C^{-1}C = I_3.$$

EXERCISE 7.13

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \in \mathbb{F}_{\le 3}[x], \qquad \ell_1(p) = p(0), \quad \ell_2(p) = p(-1), \quad \ell_3(p) = p'(0).$$

Please find the matrix of the three maps in the basis $(1, x, x^2, x^3)$, row-space and column-space bases, and the rank. Please find all p satisfying $p(0) = 2$, $p(-1) = 3$, $p'(0) = 1$, and the dimension of the solution set.

WORKED SOLUTION

Expanding the evaluations and derivative gives the matrix. The displayed minor is invertible over every field, so the rank is 3 and the listed rows and columns are bases. The solution set is an affine line of dimension 1. Integer constants denote the corresponding field elements, so the calculation also holds in positive characteristic.

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \det \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix} = -1.$$

$$B_{\text{row}} = \{(1,0,0,0), (1,-1,1,-1), (0,1,0,0)\}, \qquad B_{\text{col}} = \{(1,1,0)^T, (0,-1,1)^T, (0,1,0)^T\}.$$

$$a_0 = 2, \quad a_1 = 1, \quad 2 - 1 + a_2 - a_3 = 3 \implies a_2 = 2 + a_3.$$

$$p(x) = 2 + x + 2x^2 + t(x^2 + x^3), \qquad t \in \mathbb{F}.$$

EXERCISE 7.14

$$A = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{pmatrix}, \qquad z = ax_1 + bx_2 + cx_3.$$

Please find a necessary and sufficient condition for uniquely recovering $x \in \mathbb{R}^3$ from Ax and z. Please give one useful extra reading and one reading determined by the original two.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.14 CONTINUED

WORKED SOLUTION

Unique recovery means that the extra reading does not vanish on the kernel direction. Equivalently, the augmented square matrix is invertible. The first example satisfies the condition; the second is the sum of the two original readings.

$$\ker A = \operatorname{span}\{(3,-2,1)^T\}, \qquad (a,b,c)(3,-2,1)^T = 3a - 2b + c.$$

$$\det \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ a & b & c \end{pmatrix} = 3a - 2b + c.$$

$$3a - 2b + c \neq 0.$$

$$z = x_3: \quad (a,b,c) = (0,0,1), \qquad z = x_1 + x_2 - x_3: \quad (a,b,c) = (1,1,-1).$$

EXERCISE 7.15 EXAM QUESTION

1. Let $n \geq 2$ and $A, B \in M_n(\mathbb{R})$ satisfy $B \neq 0$ and $\operatorname{rank}(A) = n - 1$. Please prove that $AB = 0$ implies $\operatorname{rank}(B) = 1$.
2. Let $A, B \in M_{2031}(\mathbb{R})$ satisfy $AB = 0$. Please prove that
$$\operatorname{rank}(A) + \operatorname{rank}(B) \leq 2031.$$

WORKED SOLUTION

The key implication in both parts is

$$AB = 0 \implies \operatorname{Im}(B) \subseteq \ker A.$$

Indeed, every $y \in \operatorname{Im} B$ has the form $y = Bx$. Thus $Ay = A(Bx) = (AB)x = 0$, so $y \in \ker A$.

1. The inclusion and rank-nullity give
$$\begin{aligned} \operatorname{rank}(B) &= \dim \operatorname{Im}(B) \\ &\leq \dim \ker A \\ &= n - \operatorname{rank}(A) \\ &= n - (n-1) = 1. \end{aligned}$$
Since $B \neq 0$, at least one column of B is nonzero, so $\operatorname{rank}(B) \geq 1$. The two bounds yield $\operatorname{rank}(B) = 1$.
2. Apply the same inclusion in $\mathbb{R}^{2031}$:
$$\begin{aligned} \operatorname{rank}(B) &= \dim \operatorname{Im}(B) \\ &\leq \dim \ker A \\ &= 2031 - \operatorname{rank}(A). \end{aligned}$$
Rearranging gives
$$\operatorname{rank}(A) + \operatorname{rank}(B) \leq 2031.$$

If every output of B is annihilated by A, the entire image of B must lie in the kernel of A. The dimension of that kernel therefore bounds the rank of B.

Created by Chen Gadi

 © 2026 Chen Gadi

Additional practice

7

EXERCISE 7.16 FOUNDATION

$$A=\begin{pmatrix}1&-1&0&2\\0&0&1&3\\1&-1&1&5\end{pmatrix},\qquad b=(2,4,6)^T.$$

Please find row-space, column-space and kernel bases and please check rank-nullity. Please give a solvability condition for a general right-hand side and please find all solutions for the given b.

WORKED SOLUTION

The first two rows are independent and pivot columns are $1,3$. The kernel vectors have independent free entries in positions $2,4$. The row relation gives the solvability condition, and the first two equations prove sufficiency. The displayed solutions are a translate of the kernel.

$$R_3=R_1+R_2,\quad B_{\text{row}}=\{(1,-1,0,2),(0,0,1,3)\},\quad \operatorname{rank}A=2,$$

$$B_{\text{col}}=\{(1,0,1)^T,(0,1,1)^T\},$$

$$x_2=s,\quad x_4=t\quad\Longrightarrow\quad x_1=s-2t,\quad x_3=-3t,$$

$$B_{\ker}=\{(1,1,0,0)^T,(-2,0,-3,1)^T\},\qquad 2+2=4.$$

$$\exists x:\ Ax=b\iff b_3=b_1+b_2,$$

$$x=(2,0,4,0)^T+s(1,1,0,0)^T+t(-2,0,-3,1)^T,\qquad s,t\in\mathbb{R}.$$

EXERCISE 7.17 CHALLENGE

$$A\in\mathbb{F}^{m\times n},\quad B\in\mathbb{F}^{n\times p}.$$

$$\operatorname{rank}(AB)=\operatorname{rank}B-\dim(\operatorname{Col}B\cap\ker A).$$

$$A_0=\begin{pmatrix}1&0&-2\\0&1&1\end{pmatrix},\qquad B_0=\begin{pmatrix}2&0\\0&1\\1&0\end{pmatrix}\quad\text{over }\mathbb{R}.$$

Please prove the identity and please deduce $\operatorname{rank}(AB)\geq\operatorname{rank}A+\operatorname{rank}B-n$. Please verify the identity for A_0,B_0.

WORKED SOLUTION

Apply rank-nullity to the restriction L. Its image and kernel are exactly the spaces in the identity. In the example, the kernel of A_0 lies in the column space of B_0, so their intersection has dimension 1.

$$L:\operatorname{Col}B\to\mathbb{F}^m,\qquad L(y)=Ay,\quad \operatorname{Im}L=\operatorname{Col}(AB),\quad \ker L=\operatorname{Col}B\cap\ker A.$$

$$\operatorname{rank}B=\dim\operatorname{Im}L+\dim\ker L,\qquad \dim\ker L\leq\dim\ker A=n-\operatorname{rank}A.$$

$$\ker A_0=\operatorname{span}\{(2,-1,1)^T\},\quad (2,-1,1)^T=B_0e_1-B_0e_2,\quad \operatorname{rank}B_0=2,$$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.17 CHALLENGE CONTINUED

$$A_0 B_0 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \text{rank}(A_0 B_0) = 1 = 2 - 1.$$

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.18 CHALLENGE

Let $A \in M_n(F)$ over any field. Please prove

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) \geq n,$$

with equality if and only if $A^2 = A$.

WORKED SOLUTION

Put $U = \operatorname{Col} A$ and $W = \operatorname{Col}(I_n - A)$. Since $x = Ax + (I_n - A)x$, we have $U + W = F^n$. The dimension formula gives

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) = n + \dim(U \cap W) \geq n.$$

If equality holds, then $U \cap W = \{0\}$. For every x, the vector $A(I_n - A)x = (I_n - A)Ax$ belongs to both spaces, so it is zero. Hence $A^2 = A$. Conversely, assume $A^2 = A$ and let $y = Au = (I_n - A)v$. The first representation gives $Ay = y$, while the second gives $Ay = (A - A^2)v = 0$. Therefore $y = 0$, the intersection is trivial, and equality follows.

Created by Chen Gadi

 © 2026 Chen Gadi

EXERCISE 7.19 CHALLENGE

Let $m, n \geq 2$ and let $W \neq \{0\}$ be a subspace of $M_{m\times n}(F)$ in which every matrix has rank at most 1. Please prove that at least one of the following holds:

1. There is a fixed $u \in F^m \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $v \in F^n$.
2. There is a fixed $v \in F^n \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $u \in F^m$.

WORKED SOLUTION

Every nonzero rank-one matrix is uv^T with $u, v \neq 0$: choose a nonzero column u and collect the scalar multiples expressing the other columns into v^T. First consider uv^T and rs^T. If both u, r and v, s are independent pairs, the matrix with rows v^T, s^T has rank 2 and maps onto F^2. Thus there are x, y with

$$v^T x = 1,\ s^T x = 0, \qquad v^T y = 0,\ s^T y = 1.$$

The sum sends x to u and y to r, so its rank is at least 2. Consequently a sum of rank at most 1 forces dependence of at least one pair of factors. Choose $A_0 = u_0 v_0^T \in W$ nonzero. If every column space lies in span$\{u_0\}$, the first alternative holds. Otherwise choose B whose column direction is independent of u_0. Since $A_0 + B \in W$, the preceding argument forces the row factor of B to be a multiple of v_0. Absorbing that nonzero scalar into its column factor gives $B = rv_0^T$ with r, u_0 independent. For any nonzero $C = cd^T \in W$, suppose d, v_0 were independent. Applying the same argument to $A_0 + C$ forces $c \in$ span$\{u_0\}$; applying it to $B + C$ forces $c \in$ span$\{r\}$. This gives $c = 0$, a contradiction. Every row factor is therefore a multiple of v_0, proving the second alternative.

Created by Chen Gadi

 © 2026 Chen Gadi